# AppWorld

A benchmark for coding agents:
a fake world, and a dataset of tasks in that world.

## A coding agent

A program that writes and runs code to carry out a request. In AppWorld, a language model writes the next Python code.

# The world and the dataset

## The world

About 100 people. Their accounts, contacts, messages, and payments are already there before the agent starts.

Nine apps: Gmail, Venmo, Phone, Spotify, Amazon, Todoist, Splitwise, Simple Note, and a file system. A function is one Python call on an app, such as listing payments. There are 457.

## The dataset

One task is one request from one person.

It includes the request, that person's starting world, and tests that check whether the request was carried out.

## Why "fake"

The people and apps exist only inside AppWorld. The accounts and payments are not a real phone or a real Venmo.

# Two kinds of Python

## Already in the apps

The 457 functions are written in advance. The language model does not write them.

One of them is `apis.venmo.show_transactions`, which lists payments.

## Written while the task runs

The language model writes Python that calls those functions. For example:

```
apis.phone.search_contacts(...)
```

```
apis.venmo.show_transactions(...)
```

# An example of a task

*How much money have I sent to my roommates on Venmo since January 1?*

The model writes Python that calls the apps. For this task it looks like this:

```
roommates = apis.phone.search_contacts(
   access_token=phone_access_token,
   relationship="roommate")
payments = apis.venmo.show_transactions(
   access_token=venmo_access_token)
```

**How a task is scored**

AppWorld runs the tests that came with the task. The task passes when every test passes.